

# Continuous exposures and g-computation

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Normal regression estimates associations. But we want *causal* estimates: what would happen if *everyone* in the study were exposed to x vs if *no one* was exposed.

# G-Computation/G-Formula

- 1 Fit a model for  $y \sim x + z$  where  $z$  is all covariates
- 2 Create a duplicate of your data set for each level of  $x$
- 3 Set the value of  $x$  to a single value for each cloned data set (e.g  $x = 1$  for one,  $x = 0$  for the other)

# G-Computation/G-Formula

- 1 Make predictions using the model on the cloned data sets
- 2 Calculate the estimate you want, e.g.  $\text{mean}(x_1) - \text{mean}(x_0)$

# ***Advantages of the parametric G-formula***

Often more statistically precise than propensity-based methods

Incredibly flexible

Basis of other important causal models, e.g. causal survival analysis and TMLE

# Greek Pantheon data (greek\_data)

| The name of a Greek god | A prognostic factor | The treatment, a heart transplant | The outcome, death |
|-------------------------|---------------------|-----------------------------------|--------------------|
| Rheia                   | 0                   | 0                                 | 0                  |
| Kronos                  | 0                   | 0                                 | 1                  |
| Demeter                 | 0                   | 0                                 | 0                  |
| Hades                   | 0                   | 0                                 | 0                  |
| Hestia                  | 0                   | 1                                 | 0                  |
| Poseidon                | 0                   | 1                                 | 0                  |
| Hera                    | 0                   | 1                                 | 0                  |
| Zeus                    | 0                   | 1                                 | 1                  |
| Artemis                 | 1                   | 0                                 | 1                  |
| Apollo                  | 1                   | 0                                 | 1                  |

+ 10 more rows

# 1. Fit a model for $y \sim a + 1$

```
1 greek_model <- lm(y ~ a + 1, data = greek_data)
```

## 2. Create a duplicate of your data set for each level of **a**

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|-------------------------|---------------------|-----------------------------------|--------------------|
| Rheia                   | 0                   | 0                                 | 0                  |
| Kronos                  | 0                   | 0                                 | 1                  |
| Demeter                 | 0                   | 0                                 | 0                  |
| Hades                   | 0                   | 0                                 | 0                  |
| Hestia                  | 0                   | 1                                 | 0                  |
| Poseidon                | 0                   | 1                                 | 0                  |
| Hera                    | 0                   | 1                                 | 0                  |
| Zeus                    | 0                   | 1                                 | 1                  |
| Artemis                 | 1                   | 0                                 | 1                  |
| Apollo                  | 1                   | 0                                 | 1                  |



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| Hestia                  | 0                   | 1                                 | 0                  |
| Poseidon                | 0                   | 1                                 | 0                  |
| Hera                    | 0                   | 1                                 | 0                  |
| Zeus                    | 0                   | 1                                 | 1                  |
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| Poseidon                | 0                   | 1                                 | 0                  |
| Hera                    | 0                   | 1                                 | 0                  |
| Zeus                    | 0                   | 1                                 | 1                  |
| Artemis                 | 1                   | 0                                 | 1                  |
| Apollo                  | 1                   | 0                                 | 1                  |

# 3. Set the value of **a** to a single value for each cloned data set

| The name of a Greek god | A prognostic factor | a | The outcome, death |
|-------------------------|---------------------|---|--------------------|
| Rheia                   | 0                   | 0 | 0                  |
| Kronos                  | 0                   | 0 | 1                  |
| Demeter                 | 0                   | 0 | 0                  |
| Hades                   | 0                   | 0 | 0                  |
| Hestia                  | 0                   | 0 | 0                  |
| Poseidon                | 0                   | 0 | 0                  |
| Hera                    | 0                   | 0 | 0                  |
| Zeus                    | 0                   | 0 | 1                  |
| Artemis                 | 1                   | 0 | 1                  |
| Apollo                  | 1                   | 0 | 1                  |

| The name of a Greek god | A prognostic factor | a | The outcome, death |
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| Hades                   | 0                   | 1 | 0                  |
| Hestia                  | 0                   | 1 | 0                  |
| Poseidon                | 0                   | 1 | 0                  |
| Hera                    | 0                   | 1 | 0                  |
| Zeus                    | 0                   | 1 | 1                  |
| Artemis                 | 1                   | 1 | 1                  |
| Apollo                  | 1                   | 1 | 1                  |

### 3. Set the value of **a** to a single value for each cloned data set

```
1 # set all participants to have a = 0
2 untreated_data <- greek_data |>
3   mutate(a = 0)
4
5 # set all participants to have a = 1
6 treated_data <- greek_data |>
7   mutate(a = 1)
```

## 4. Make predictions using the model on the cloned data sets

```
1 # predict under the data where everyone is untreated
2 predicted_untreated <- greek_model |>
3   augment(newdata = untreated_data) |>
4   select(untreated = .fitted)
5
6 # predict under the data where everyone is treated
7 predicted_treated <- greek_model |>
8   augment(newdata = treated_data) |>
9   select(treated = .fitted)
10
11 predictions <- bind_cols(
12   predicted_untreated,
13   predicted_treated
14 )
```

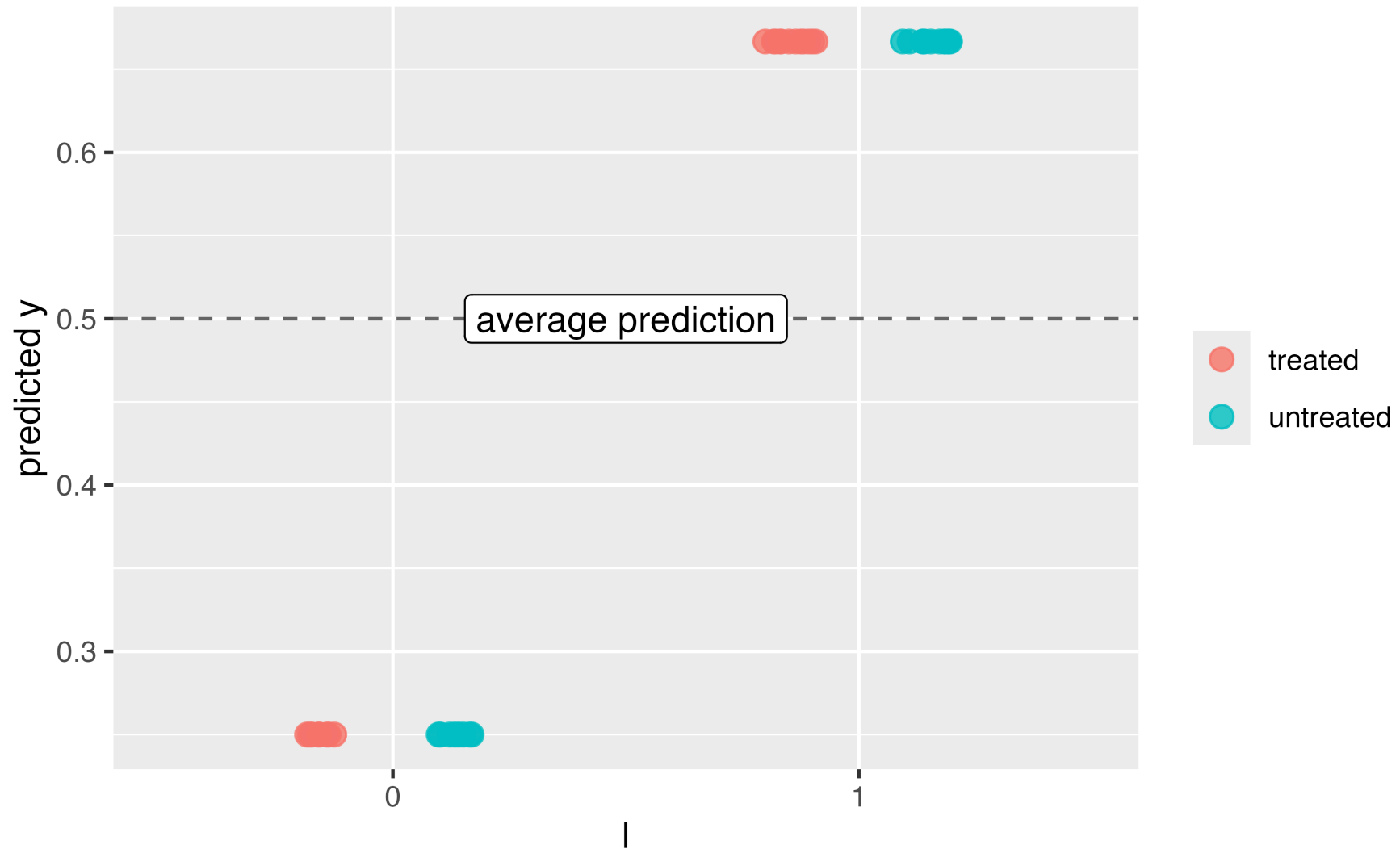
# 5. Calculate the estimate you want

```
1 predictions |>
2   summarise(
3     mean_treated = mean(treated),
4     mean_untreated = mean(untreated),
5     difference = mean_treated - mean_untreated
6   )
```

# A tibble: 1 × 3

|   | mean_treated | mean_untreated | difference |
|---|--------------|----------------|------------|
|   | <dbl>        | <dbl>          | <dbl>      |
| 1 | 0.5          | 0.5            | 0          |

# Marginalizing over the covariates



Predict at every observation's actual covariates, then average

# avg\_comparisons() does all five steps

```
1 library(marginaleffects)
2
3 avg_comparisons(greek_model, variables = "a")
```

```
# A tibble: 1 × 4
  estimate std.error conf.low conf.high
  <dbl>    <dbl>    <dbl>    <dbl>
1      0      0.24   -0.47     0.47
```

The standard error comes from the **delta method**, which approximates the variance of a transformation of the estimated coefficients

# Targeting estimands with **newdata**

**newdata** changes whose covariates we average over

```
1  # ATT: average over the treated
2  avg_comparisons(
3    greek_model,
4    variables = "a",
5    newdata = filter(greek_data, a == 1)
6  )
7
8  # ATC: average over the untreated
9  avg_comparisons(
10   greek_model,
11   variables = "a",
12   newdata = filter(greek_data, a == 0)
13 )
```



# Same model, three populations

```
# A tibble: 3 × 4
  estimand estimate conf.low conf.high
  <chr>      <dbl>    <dbl>    <dbl>
1 ATE          0    -0.47     0.47
2 ATT          0    -0.47     0.47
3 ATC          0    -0.47     0.47
```

Without an exposure-covariate interaction, the three coincide: the model assumes a constant effect across covariate values

# Tilting toward a target population

The ATO and ATM populations are not subsets of the rows, so weight the average by the tilting function instead

```
1 library(propensity)
2
3 greek_ps <- glm(a ~ 1, data = greek_data, family = binomial()) |>
4   predict(type = "response")
5
6 avg_comparisons(
7   greek_model,
8   variables = "a",
9   wts = ps_tilt(greek_ps, "ato")
10 )
```

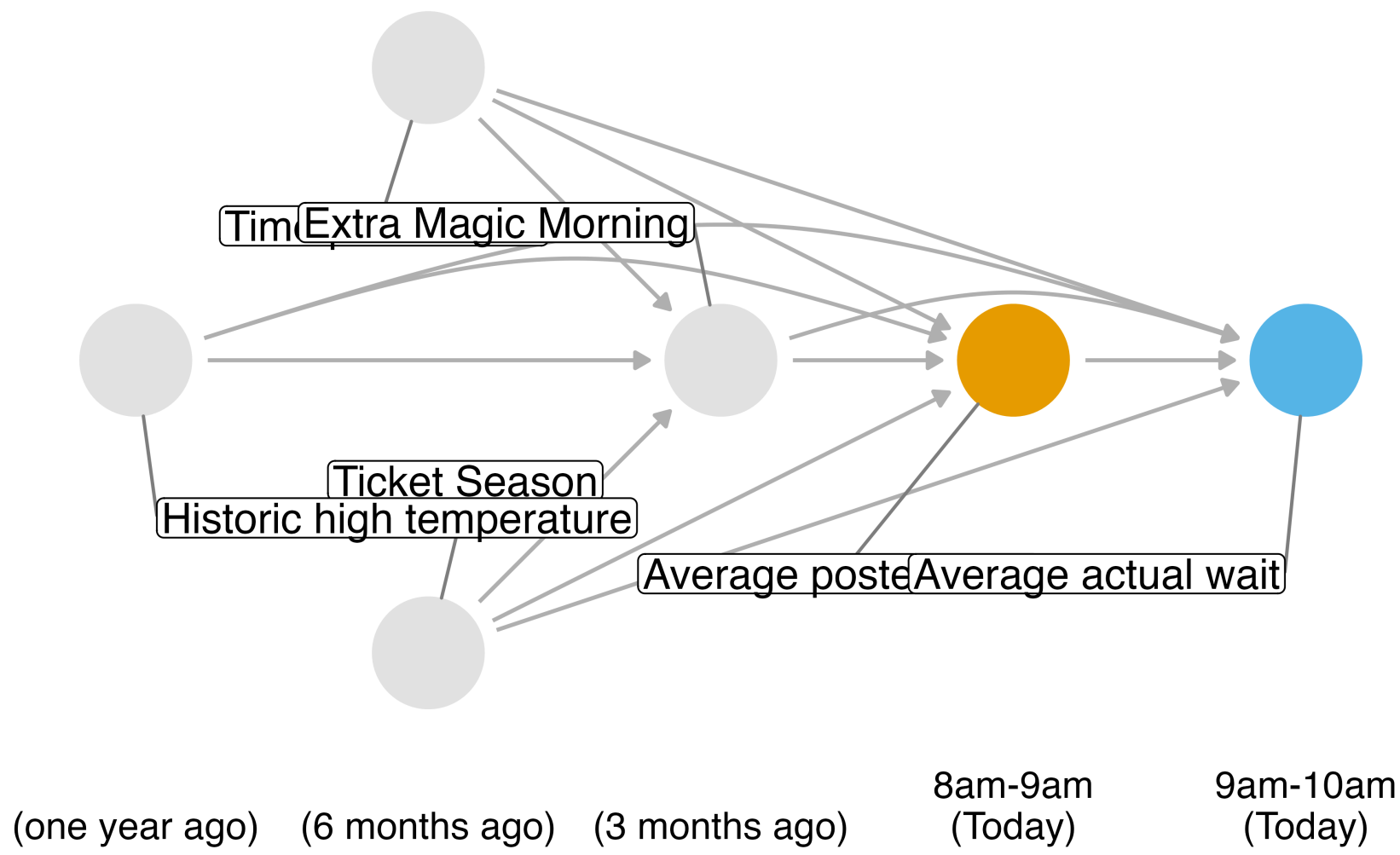
# Continuous exposures

**We recommend g-computation  
over propensity scores for  
continuous exposures because of  
stability issues**

When overlap is poor, IPW makes its reliance on overlap explicit: the weights blow up and we are forced to confront the problem.

G-computation absorbs the problem silently, producing a stable estimate that may nonetheless be wrong.

# Do *posted* wait times at 8 am affect *actual* wait times at 9 am?



## ***Your Turn***

**Work through Your Turns 1-4 in `11-continuous-g-computation-exercises.qmd`**